

Circles — MCQ Worksheet

Mathematics • Chapter 10 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. A line that touches a circle at exactly one point is called a:

- | | |
|-------------|--------------|
| (A) Chord | (B) Secant |
| (C) Tangent | (D) Diameter |

Q2. A line that intersects a circle in two distinct points is called a:

- | | |
|-------------|------------|
| (A) Tangent | (B) Secant |
| (C) Chord | (D) Radius |

Q3. The tangent to a circle is perpendicular to the radius at the:

- | | |
|--------------|----------------------|
| (A) Centre | (B) Point of contact |
| (C) Midpoint | (D) Diameter |

Q4. The number of tangents that can be drawn from an external point to a circle is:

- | | |
|-------|-------|
| (A) 0 | (B) 1 |
| (C) 2 | (D) 3 |

Q5. The number of tangents that can be drawn from a point on the circle is:

- | | |
|-------|--------------|
| (A) 0 | (B) 1 |
| (C) 2 | (D) Infinite |

Q6. The number of tangents that can be drawn from a point inside the circle is:

- | | |
|-------|--------------|
| (A) 0 | (B) 1 |
| (C) 2 | (D) Infinite |

Q7. Lengths of tangents drawn from an external point to a circle are:

- | | |
|--------------|-------------|
| (A) Equal | (B) Unequal |
| (C) One zero | (D) Random |

Q8. If a tangent of length 8 cm is drawn to a circle of radius 6 cm from an external point, the distance of the point from centre is:

- | | |
|-----------|-----------|
| (A) 10 cm | (B) 8 cm |
| (C) 6 cm | (D) 12 cm |

Q9. If two tangents from an external point P make an angle of 60° between them and $OP = 10$, the radius of circle is:

- | | |
|----------------------|--------------------|
| (A) 5 cm | (B) $5\sqrt{3}$ cm |
| (C) $10/\sqrt{3}$ cm | (D) 10 cm |

Q10. Tangent at any point of a circle is perpendicular to:

- | | |
|--------------------------------|---------------------|
| (A) Diameter | (B) Chord |
| (C) Radius drawn at that point | (D) Another tangent |

Q11. The angle between two tangents drawn from an external point is supplementary to the angle subtended by the line segment joining points of contact at the centre. This means the sum is:

- | | |
|-----------------|-----------------|
| (A) 90° | (B) 180° |
| (C) 360° | (D) 270° |

Q12. In a circle of radius 5 cm, the length of a chord at a perpendicular distance of 3 cm from the centre is:

- | | |
|-----------|----------|
| (A) 8 cm | (B) 6 cm |
| (C) 10 cm | (D) 4 cm |

Q13. The tangent length from an external point at distance 13 from centre to a circle of radius 5 is:

- | | |
|--------|--------|
| (A) 12 | (B) 13 |
| (C) 8 | (D) 18 |

Q14. If tangents PA and PB from a point P touch a circle at A and B, then OAPB is a:

- | | |
|--------------------------|---------------|
| (A) Square | (B) Rectangle |
| (C) Cyclic quadrilateral | (D) Trapezium |

Q15. The perpendicular from the centre to a chord:

- | | |
|--------------------------------|--------------------------|
| (A) Bisects the chord | (B) Is parallel to chord |
| (C) Is at midpoint of arc only | (D) Equals chord |

Q16. Two parallel tangents to a circle are at a distance equal to:

- | | |
|---------------------|------------------------|
| (A) Radius | (B) Diameter |
| (C) Half the radius | (D) Twice the diameter |

Q17. In two concentric circles of radii 5 cm and 3 cm, the length of chord of larger circle which touches the smaller is:

- | | |
|----------|-----------|
| (A) 4 cm | (B) 6 cm |
| (C) 8 cm | (D) 10 cm |

Q18. The angle in a semicircle is:

- | | |
|------------|--------------------------|
| (A) Acute | (B) Right (90°) |
| (C) Obtuse | (D) Reflex |

Q19. PQ is a chord of a circle and the tangent at P makes an angle of 50° with PQ. Then the angle subtended by PQ at the centre is:

- | | |
|----------------|-----------------|
| (A) 50° | (B) 100° |
| (C) 80° | (D) 130° |

Q20. If two tangents from external point are drawn to a circle and the line from external point to centre bisects the angle between tangents, this is true:

- | | |
|------------|---------------------|
| (A) Always | (B) Sometimes |
| (C) Never | (D) Only for square |

Q21. The tangent and radius at the point of contact form an angle of:

- | | |
|----------------|----------------|
| (A) 45° | (B) 60° |
| (C) 90° | (D) 30° |

Q22. In a quadrilateral ABCD circumscribing a circle, $AB + CD =$

- | | |
|--------------------|---------------|
| (A) $AC + BD$ | (B) $AD + BC$ |
| (C) $AC \times BD$ | (D) $AC - BD$ |

Q23. From a point Q at distance 5 cm from the centre of a circle of radius 3 cm, the length of tangent is:

- | | |
|----------|--------------------|
| (A) 4 cm | (B) 5 cm |
| (C) 8 cm | (D) $\sqrt{34}$ cm |

Q24. PA and PB are two tangents drawn to a circle at points A and B from external point P. If angle $APB = 70^\circ$, then $\angle AOB =$

- | | |
|-----------------|-----------------|
| (A) 110° | (B) 70° |
| (C) 140° | (D) 100° |

Q25. The longest chord of a circle is the:

- | | |
|-------------|--------------|
| (A) Radius | (B) Diameter |
| (C) Tangent | (D) Sector |